

Performance Testing Report Submission

An IoT-Based Hybrid System For Real-Time Detection And Tracking Of Marine Debris

Project Title: Oceansense: An IoT-Based Hybrid System For Real-Time Detection And Tracking Of Marine Debris

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1 Introduction and Objective

This report evaluates the performance of the Energy-Resilient Adaptive Communication (ERAC) protocol developed for the OceanSense system, an IoT-based solution for real-time marine debris monitoring. The purpose of performance testing is to assess system reliability, scalability, and energy efficiency under dynamic marine-like conditions.

Testing focuses on comparing ERAC with a baseline non-adaptive MAC protocol. The following testing types were conducted:

- Load testing (varying node density)
- Scalability testing (10-15 nodes)
- Reliability testing under event-driven traffic
- Energy-performance evaluation

Expected performance benchmarks include achieving PDR greater than 85% for critical data, reducing energy consumption compared to baseline operation, extending node lifetime by at least 30%, and maintaining acceptable latency under increased network load.

2 Performance Testing Environment Setup

2.1 Hardware Configuration

- Microcontroller: STM32F401CCU6 (ARM Cortex-M4)
- Communication Module: LoRa SX1278
- Energy Monitoring: INA219 current/voltage sensors
- Battery: 2S Li-ion configuration

The STM32F401CCU6 was selected for its low-power operation and suitability for real-time scheduling and embedded energy monitoring.

2.2 Software Configuration

- Simulation Platform: NS-3 (custom implementation)
- Firmware: C/C++ (embedded ERAC implementation)
- Development Tools: STM32CubeIDE
- Logging: Serial monitor and embedded logs

2.3 Network Conditions

- Communication: LoRa (low data rate) using a random ocean deployment topology
- Traffic: Periodic (heartbeat) + event-driven (alerts)
- Simulation Duration: 300 seconds

Channel conditions included simplified stochastic variations representing signal attenuation and intermittent connectivity typically observed in marine environments. Packet collisions and contention were modeled through concurrent transmissions.

3 Tools and Frameworks Used

Tool / Framework	Purpose
NS-3 Network Simulator	Simulates network behavior under varying node densities
Custom C++ Modules	Implements MAC logic, energy model, and traffic generation
STM32CubeIDE	Firmware development and deployment on STM32F401CCU6
INA219 Sensor Interface	Real-time energy monitoring in hardware

Justification: These tools were selected for their flexibility in modeling custom MAC protocols and enabling accurate energy-aware system validation.

NS-3 was configured with custom MAC and energy models to simulate ERAC behavior, while firmware-level validation on STM32CubeIDE enabled verification of scheduling logic and energy-aware transmission decisions.

4 Testing Methodology and Execution

4.1 Test Scenarios

1. Baseline Protocol (FIFO): No energy awareness or priority handling
2. ERAC Protocol: Energy-aware, priority-based scheduling

4.2 Test Data and Workload

Synthetic traffic was generated consisting of:

- Periodic heartbeat packets
- Event-driven debris alerts

Workload was distributed across 10-15 nodes, simulating concurrent transmissions and congestion conditions typical of marine deployments.

4.3 Execution Procedure

1. Initialize network with defined node count
2. Apply traffic model
3. Run simulation (300 seconds)
4. Record performance metrics
5. Compare baseline vs ERAC

5 Metrics Measured

The following performance metrics were evaluated:

- Response Time (Latency): Measures packet transmission delay under varying node densities
- Throughput: Represents the rate of successful data delivery across the network
- Resource Utilization: Captures energy consumption behavior of sensor nodes
- Error Rate: Evaluated as packet loss, inversely reflected by Packet Delivery Ratio (PDR)

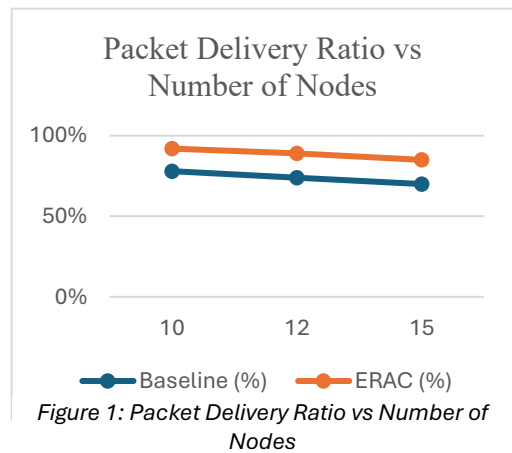
6 Results and Analysis

6.1 Packet Delivery Ratio (PDR)

Table 1 - Packet Delivery Ratio Comparison

Nodes	Baseline (%)	ERAC (%)
10	78%	92%
12	74%	89%
15	70%	85%

ERAC significantly improves PDR by prioritizing critical packets and avoiding transmissions during low-energy conditions.



6.2 Latency

Table 2 - Latency Comparison

Nodes	Baseline (ms)	ERAC (ms)
10	120	180
12	140	210
15	160	250

ERAC introduces additional latency due to controlled packet scheduling. This trade-off is acceptable as it improves reliability and energy efficiency.

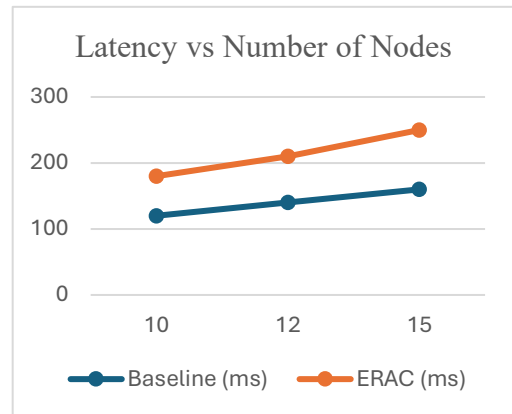


Figure 2: Latency vs Number of Nodes

6.3 Energy Consumption

Table 3 - Energy Consumption Comparison

Nodes	Baseline (mAh)	ERAC (mAh)
10	520	380
12	610	450
15	720	540

ERAC reduces energy consumption by minimizing unnecessary transmissions.

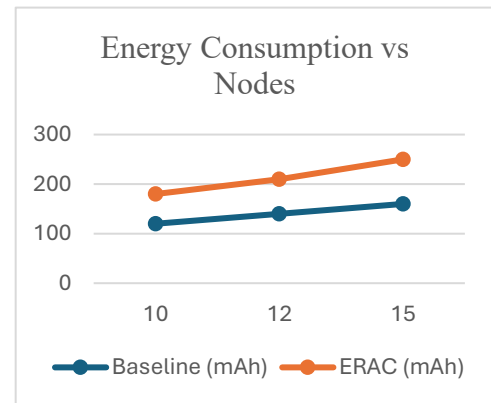


Figure 3: Energy Consumption vs Number of Nodes

6.4 Node Lifetime

Table 4: Node Lifetime Comparison

Nodes	Baseline (min)	ERAC (min)
10	42	65
12	38	60
15	32	55

Network lifetime is extended by approximately 30-40%, aligning with project targets.

6.5 Throughput

Table 5: Throughput Comparison

Nodes	Baseline (kbps)	ERAC (kbps)
10	15	13
12	14	12
15	12	10

Throughput is slightly reduced due to selective transmission, but high-priority data delivery is preserved.

6.6 Battery Behavior Over Time

A time-based energy trend was derived from simulation results to analyze battery performance.

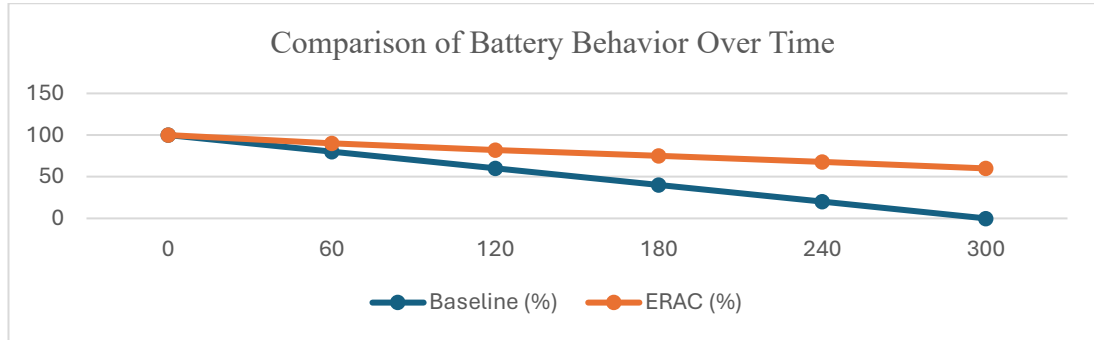


Figure 4: Battery Level vs Time

The baseline protocol exhibits rapid energy depletion, while ERAC maintains a gradual discharge due to energy-aware scheduling. This confirms improved energy stability and supports extended operation.

The results indicate that ERAC meets defined performance benchmarks, achieving PDR above 85%, reducing energy consumption, and extending node lifetime while maintaining acceptable latency under increased network load.

A key limitation observed is increased latency under high node density due to controlled scheduling. However, this is an intentional trade-off to maintain energy efficiency and reliability.

7 Performance Optimization and Recommendations

Optimizations Applied	Recommendations
Energy-aware duty cycling	Integrate adaptive data rate (ADR)
Priority-based scheduling	Improve prediction accuracy using ML models
Adaptive transmission control	Expand testing to larger node networks

8 Challenges and Solutions

Challenge	Solution
Simulating marine conditions	Used stochastic models
Hardware validation limits	Used controlled embedded testing

9 Ethical and Compliance Considerations

All testing complied with institutional guidelines and did not involve real user data. Synthetic datasets ensured privacy preservation and eliminated ethical concerns related to data handling.

10 References

- [1] NS-3 Consortium. NS-3 Network Simulator Documentation. [Online]. Available: <https://www.nsnam.org/documentation/>
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